

Project Design Phase Proposed Solution Template

Date	26 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No	Parameter	Description
1	Problem Statement (Problem to be Solved)	Governments, researchers, and policymakers face challenges in evaluating and comparing the Human Development Index (HDI) of countries because it requires analyzing multiple socio-economic indicators such as Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and Gross National Income (GNI) per capita. Manual analysis is time-consuming, complex, and may lead to inconsistent conclusions. There is a need for a faster, accurate, and data-driven HDI prediction system.
2	Idea / Solution Description	The Human Development Index (HDI) Predictor is a Machine Learning-powered web application that predicts the HDI score of a country using key development indicators such as Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per Capita. The system performs data preprocessing, feature selection, and prediction using a Linear Regression model. The trained model is integrated into a Flask web application to provide real-time HDI prediction along with classification into Very High, High, Medium, or Low human development categories.
3	Novelty / Uniqueness	The proposed solution combines machine learning, statistical analysis, and an interactive Flask web application to provide instant HDI prediction and classification. It enables users to understand how socio-economic indicators influence human development while offering a simple, user-friendly interface for prediction, visualization, and comparative analysis.
4	Social Impact / Customer Satisfaction	The system assists governments, researchers, students, and international organizations in understanding development levels more efficiently. It supports evidence-based policy formulation, identifies development gaps, promotes transparency in socio-economic analysis, and reduces the effort required for manual HDI evaluation. This contributes to improved public policy planning, educational research, and sustainable development initiatives.
5	Business Model (Revenue Model)	The application can be offered as a Software-as-a-Service (SaaS) platform to government agencies, educational institutions, research organizations, NGOs, and international development organizations through subscription-based licensing, institutional deployment, or API-based prediction services. Premium analytics dashboards and country comparison reports can also be provided as value-added services.
6	Scalability of the Solution	The application can be deployed on cloud platforms such as IBM Cloud, Render, Railway, or AWS and scaled to support thousands of prediction requests. Future enhancements may include additional machine learning algorithms, historical trend analysis, AI-based policy recommendation systems, interactive dashboards, and integration with global development datasets (e.g., UNDP or World Bank) without significant architectural changes.